# Open-Vocabulary 3D Scene Graphs from Sparse Views:
## Duplicate-Aware Fusion and Honest Evaluation

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**Abstract.** Open-vocabulary 3D scene graphs, labeled objects and the spatial relations among them, are useful for embodied and mixed-reality agents, yet most systems assume posed RGB-D video or a prebuilt 3D map. We target the harder regime of a handful of casually captured RGB views, and argue that progress here depends as much on honest evaluation as on the pipeline itself. Our system keeps every heavy model frozen and uses no per-scene optimization: feed-forward VGGT geometry turns the sparse images into world-point maps, an open-vocabulary detector (OWLv2) proposes labeled object boxes, and these are lifted to 3D and fused across views into an object–relation graph. A genuine detector front-end is what makes the labels real: the common class-agnostic-mask plus per-patch CLIP alternative returns scene-scale category noise on casual frames. We evaluate against an independently authored reference, built from the frames rather than seeded from the model's own predictions. The dominant error is then *over-counting*, one physical object surviving as several fused nodes, and a one-line duplicate-aware merge corrects it, improving object-label F1 at every view count and winning on 22–27 of 28 TUM RGB-D scenes ($p < 0.001$). By contrast, a previously promising uncertainty-aware fusion gate gives no benefit across a full weight sweep, a negative result that the earlier prediction-seeded, circular reference had masked. We release the benchmark and the dataset-agnostic de-circularization protocol.

**Keywords:** 3D scene graphs · sparse views · open-vocabulary detection · evaluation methodology · benchmark · duplicate-aware fusion

## 1 Introduction

Reconstructing a scene and explaining a scene differ. Feed-forward geometry models recover camera motion, depth, and point maps from a few images [29,13,28], and 2D foundation models give open-vocabulary detections, masks, and descriptors [17,22,11,19]. Many applications instead need object-level structure: which pixels across views belong to the same object, where it sits in 3D, and how it relates to others, so a robot can act on "the monitor is on the desk" rather than a raw point cloud. These components do not provide this alone, and the gap is widest for a handful of casual RGB views rather than a calibrated RGB-D stream or prebuilt map.

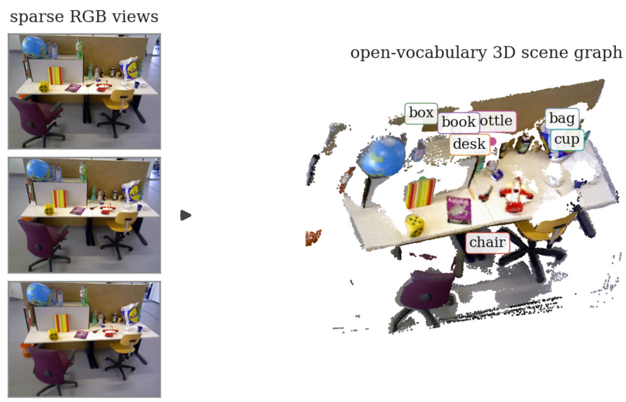


**Fig. 1.** From a handful of RGB views to an open-vocabulary 3D scene graph. **Left:** three input frames (TUM `freiburg3_long_office_household`). **Right:** frozen-model detections lifted into VGGT world-point geometry and fused across views into labeled 3D object nodes (colored markers), with all heavy models frozen and no per-scene optimization. Best viewed in color.

We build the scene graph by fusing lifted 2D detections, keeping every heavy model frozen with no per-scene optimization. VGGT world-point maps act as a 2D-to-3D lookup, an open-vocabulary detector proposes labeled object boxes, and we lift and fuse these across views into an object–relation graph (Figure 1). Fixed perception isolates the graph-building stage for ablation and places us in a lower-input regime than open-vocabulary 3D systems assuming posed RGB-D video or a dense map [5,7].

A central lesson of this work concerns evaluation. An earlier version graded predicted graphs against a reference seeded from the model's own dense-view output, which is circular: a method reproducing its ten-view graph scores 1.0 at the reference view by construction, rewarding agreement with the model rather than the scene. Rebuilding the reference from the frames themselves changes two conclusions. First, a mask-plus-CLIP front-end recovers almost no real objects; on a desk scene it returns "floor", "curtain", and "bed" instead of the present monitors and keyboard, whereas a genuine open-vocabulary detector (OWLv2) reads the actual objects. Second, an uncertainty-aware fusion gate that looked promising under the old protocol gives no improvement once labels are real, across a full weight sweep. With honest labels, the dominant error is over-counting: one physical object survives fusion as several same-label nodes (Figure 5). A one-line merge of same-label nodes with overlapping 3D boxes removes most of it, the only variant whose accuracy improves with views.

We test these on 28 TUM RGB-D sequences [25] at 3/5/8/10 views, against a reference authored from the frames (5 scenes human-verified, 23 drafted in two passes). Duplicate-aware fusion improves object-label F1 over the graph-fusion baseline by 0.06 to 0.12 and wins on 22 to 27 of the 28 scenes ($p < 0.001$), while the uncertainty gate is consistently worse. We make four contributions. **(1)** A detector-grounded open-vocabulary scene-graph system for sparse views that keeps OWLv2 and VGGT frozen with no per-scene optimization, recovering real objects a mask-plus-CLIP front-end misses. **(2)** Duplicate-aware cross-view fusion, a single post-fusion merge of same-label nodes with overlapping 3D boxes that targets over-counting and is the only variant whose accuracy improves with more views. **(3)** An independent, de-circularized benchmark on TUM RGB-D, showing how a prediction-seeded reference inflates scores while an independently authored one removes the effect. **(4)** A negative result on uncertainty-aware fusion, with analysis: the gate raises recall but over-splits, costing more precision than it gains, so merging helps.

## 2 Related Work

*Feed-forward sparse-view geometry.* Classical geometry recovery rests on correspondence, triangulation, and bundle adjustment over calibrated or densely overlapping views, as in COLMAP [23] and ORB-SLAM3 [2]; both are accurate but need many views or per-scene optimization. A recent line replaces that optimization with one network pass: DUSt3R regresses point maps from an image pair [29], MASt3R adds 3D-aware matching [13], Fast3R extends it to many views [31], and VGGT predicts cameras, depth, point maps, and per-pixel confidence from one or many views at once [28]. We treat VGGT as a *frozen* source of pixel-aligned 3D evidence; our output is an object–relation graph, not a reconstruction.

*Open-vocabulary detection as a labeling front-end.* A lifted scene graph is only as good as its 2D labels. The common recipe pairs class-agnostic SAM masks [11] with per-region CLIP embeddings [22], sometimes adding DINOv2 [19]; it is frozen but hands naming to a whole-image CLIP model applied to crops, which returns scene-level guesses rather than the object inside the crop. Open-vocabulary *detectors* instead localize and name in one pass: OWL-ViT [18] and its scaled successor OWLv2 [17], Grounding DINO [15], Detic [32], and the real-time YOLO-World [3]. We build our front-end on OWLv2 because it emits localized, named instances, exactly what lifting and fusion consume, and we measure what that buys a sparse-view 3D pipeline over the mask-then-embed alternative.

*Open-vocabulary 3D semantics.* A large body of work lifts or distills foundation-model features into points, maps, or neural representations. Building on NeRF [16], LERF embeds language in a radiance field [10], while OpenScene [20], ConceptFusion [7], OpenMask3D [26], and OpenSplat3D [21] fuse open-set features over point clouds, posed RGB-D, masks, or Gaussian splats [9]. Our goal is complementary: rather than a dense map or radiance field, we build an explicit object–relation graph from a few RGB views. CUA-O3D proposes uncertainty-aware aggregation for fusing such features across views [14]; we revisit that idea on real detector labels under an independent reference and report a negative result (Section 5).

*Open-vocabulary 3D scene graphs.* Scene graphs cast a scene as objects and their relations [1], with datasets and predictors for reconstructed indoor scenes [27,4]. Recent work makes these open-vocabulary: ConceptGraphs builds them from posed RGB-D sequences [5], Open3DSG predicts open-set objects and relations from a complete 3D point cloud [12], and HOV-SG adds a floor–room–object hierarchy for language-guided navigation [30]. All three begin from dense or posed 3D input; we share their object–relation goal but take only a handful of unposed RGB frames, lifting them through frozen feed-forward geometry. That sparse-view input makes one failure mode dominant: *over-counting*, where a single physical object survives cross-view fusion as several same-label nodes. Removing those duplicates is the job of our fusion step.

*Honest evaluation and reproducibility.* Clean instance-level ground truth is expensive for casual RGB views, so it is tempting to grade a method against a reference derived from its own dense-view output. This couples reference to predictor and yields a *circular* evaluation in which a method scores perfectly at the reference view almost by construction, a form of information leakage that drives the reproducibility crisis [8,6]. The standard response in 3D is to evaluate on controlled benchmarks with independent ground truth, such as Replica [24] and TUM RGB-D [25]. In that spirit we author a sparse-view reference directly from the frames, independently of the system under test, document the circular-pseudo-reference pitfall, and report what helps once the coupling is removed.

# 3 Method

## 3.1 Problem Setup and Overview

Given a sparse set of RGB views $\mathcal{I} = \{I_1, \dots, I_V\}$ of a static indoor scene, we build an open-vocabulary 3D scene graph $\mathcal{G} = (\mathcal{O}, \mathcal{R})$. An object node $o \in \mathcal{O}$ stores a 3D point set, centroid, axis-aligned 3D box, open-vocabulary label, supporting views, an optional appearance descriptor, and its source detections; a relation edge $r \in \mathcal{R}$ carries a spatial predicate between two nodes. The input is RGB only, with no depth, no camera poses, and no prebuilt mesh. Every heavy model is frozen and run off-the-shelf, and the stages that assemble a graph from their outputs are training-free geometric operations, so the system needs no per-scene optimization.

Figure 2 traces the flow. Feed-forward VGGT [28] gives a per-view world-point map and confidence; OWLv2 [17] gives labeled detections, so each proposal already carries a real object label rather than a region awaiting classification. We lift each detection into a shared 3D frame through the world-point map, fuse candidates across views into objects, merge over-detected copies of one object in a single duplicate-aware pass, and read spatial relations off the fused geometry. The exported graph stores each label and relation beside the evidence that produced it, so it can be inspected or turned into annotation packets.

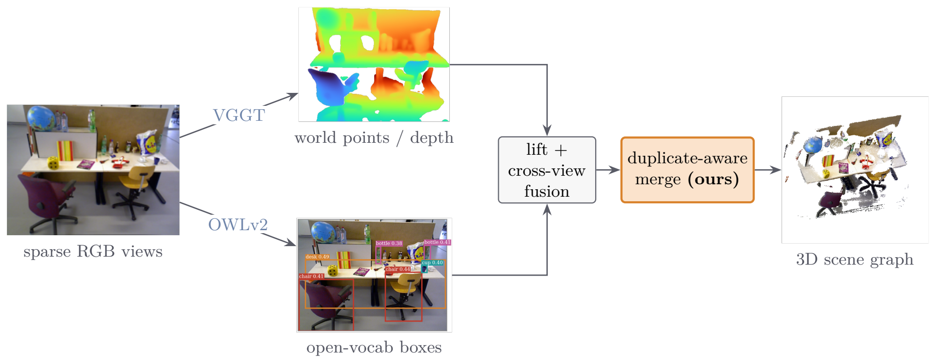


**Fig. 2.** The pipeline on a real office scene. Sparse RGB views feed two *frozen* models: VGGT produces world-point geometry (depth shown) and OWLv2 proposes open-vocabulary boxes. Detections are lifted into the world-point map and fused across views into object nodes; our one addition, a *duplicate-aware merge* (amber), collapses same-label overlapping nodes into the labeled 3D object–relation graph at right. No fine-tuning, no per-scene optimization.

### 3.2 Sparse-View Geometry

We run VGGT once on the selected views, frozen [28]. Without bundle adjustment it predicts, in one forward pass, a dense world-point map with depth, camera parameters, and per-pixel confidence; we write the world-point map as $P_v(u) \in \mathbb{R}^3$ and the confidence as $C_v(u)$ at pixel $u$ of view $v$. Only $P_v$ and $C_v$ flow downstream: the world-point map is a direct 2D-to-3D lookup placing any pixel into a common 3D frame, so we never build a mesh, run structure-from-motion, or optimize a radiance-field or Gaussian-splat representation [23,16,9]. All thresholds below are in the units of this map.

### 3.3 Open-Vocabulary Detection Front-End

On each view we run OWLv2 [17], which takes free-form text queries and returns boxes with per-box class scores. The queries are the benchmark's indoor vocabulary, so the detector itself, not a downstream patch classifier, assigns the open-vocabulary label, reading the actual objects off each frame (Fig. S1, supplementary). For detection $k$ in view $v$ we keep its box, score $s_{v,k}$, and label $\ell_{v,k}$. We use the `owlv2-base-patch16-ensemble` checkpoint at score threshold 0.2 with per-class non-maximum suppression at box-IoU 0.5, which mainly tidies large surface classes such as a desk. We keep the threshold loose on purpose: one object may receive several overlapping boxes per frame, but those boxes lift to nearly the same 3D points and collapse later (Section 3.6).

This is our one departure from the common recipe (Section 2), which pairs class-agnostic masks with whole-image CLIP on crops; here the detector box *is* the labeled object hypothesis, and everything downstream is unchanged, so any accuracy difference is attributable to the front-end. Each box optionally prompts a frozen SAM [11] for a tight mask $M_{v,k}$ so that lifting selects object pixels rather than the whole box; without a mask it falls back to the box rectangle. A separate frozen stage extracts CLIP [22] and DINOv2 [19] descriptors, stored as a normalized concatenation and used only to gate fusion and as provenance, never for labels.

### 3.4 3D Lifting

Each detection's region, its mask $M_{v,k}$ when present and its box rectangle otherwise, is rescaled to the VGGT geometry resolution and used to read world points. Non-finite points and points below the confidence threshold $\tau_c$ are dropped:

$$X_{v,k} = \{\, P_v(u) \mid u \in M_{v,k},\ C_v(u) \geq \tau_c \,\}, \qquad \tau_c = 0.2. \tag{1}$$

A detection with no surviving points is discarded. Each candidate node stores its lifted points $X_{v,k}$ (subsampled to a fixed cap), its centroid $\mu_{v,k}$, an axis-aligned 3D box, the label $\ell_{v,k}$ and score $s_{v,k}$, and, when available, the descriptor $f_{v,k}$.

### 3.5 Cross-View Fusion

Lifted candidates are assembled through an association graph. We forbid edges between two detections from the same view, so a connected component is cross-view support for one object, not a grouping within an image. Two candidates $a$ and $b$ are joined when their centroids are close and, when both carry descriptors, their appearance is similar:

$$\|\mu_a - \mu_b\|_2 \leq \epsilon_g \quad \text{and} \quad \cos(f_a, f_b) \geq \epsilon_f, \tag{2}$$

with $\epsilon_g = 0.15$ and $\epsilon_f = 0.75$; the appearance term is skipped when descriptors are absent. Each component becomes one object: its points are pooled and subsampled, its descriptors averaged and renormalized, and its label set by plurality vote over the constituent detections. This separates view-local evidence from object identity while keeping the per-detection provenance.

### 3.6 Duplicate-Aware Fusion

Cross-view fusion still leaves the scene *over-counted*, for two structural reasons: the detector over-fires (several overlapping boxes per object per frame), and same-view edges are forbidden, so duplicates within a frame are never joined. A desk or monitor thus survives as several fused nodes, the dominant precision loss against the independent reference (Section 5). Over-counting is an over-*splitting* error, so the corrective move is to merge.

We add one post-fusion pass, DEDUP, that merges same-label fused nodes whose axis-aligned 3D boxes overlap. For a node $o$ with point set $X_o$, let $B_o = [\, b_o^{\min}, b_o^{\max} \,]$ be its axis-aligned box, with $b_o^{\min}$ and $b_o^{\max}$ the coordinate-wise minimum and maximum of $X_o$. The axis-aligned 3D intersection-over-union of two boxes is

$$\mathrm{IoU}_{3D}(B_a, B_b) = \frac{\mathrm{vol}(B_a \cap B_b)}{\mathrm{vol}(B_a) + \mathrm{vol}(B_b) - \mathrm{vol}(B_a \cap B_b)}, \tag{3}$$

with $\mathrm{vol}(B_a \cap B_b) = \prod_{d=1}^{3} \max\big(0, \min(b_{a,d}^{\max}, b_{b,d}^{\max}) - \max(b_{a,d}^{\min}, b_{b,d}^{\min})\big)$. Partitioning the nodes by label, we connect every pair with $\mathrm{IoU}_{3D} > \tau$ and collapse each connected component into one object via the same fuse step:

$$\textsc{Dedup}(\mathcal{O};\tau) = \bigcup_{\ell} \Big\{ \textsc{Fuse}(c) \; : \; c \in \mathrm{CC}\big(G_\ell, \; \{(a,b) : \mathrm{IoU}_{3D}(B_a, B_b) > \tau\}\big) \Big\}, \tag{4}$$

where $G_\ell$ is the set of nodes with label $\ell$, $\mathrm{CC}(\cdot)$ denotes connected components, and $\textsc{Fuse}(\cdot)$ re-fuses a component into a single node (the identity on singletons). Unlabeled nodes and singletons pass through unchanged. The same-label restriction keeps distinct classes apart, and the box-overlap test keeps spatially separated same-class instances apart. We use 3D-box IoU rather than mask or point-set overlap because the boxes are already attached to each node, making the test parameter-free beyond $\tau$; the default $\tau = 0.1$ is robust across the low-IoU range (Section 4). Its one cost is that two genuinely distinct same-class instances with overlapping boxes, such as two monitors on one desk, can be over-merged; we quantify this and point to appearance-based instance splitting as the remedy in Sections 4 and 6.

### 3.7 Spatial Relations

Spatial relations are read off the fused centroids by a small, auditable set of distance and direction rules, one directed edge per ordered object pair. With $\delta = \mu_{\mathrm{obj}} - \mu_{\mathrm{subj}}$, a pair closer than 0.35 is labeled *near*, which preempts the directional predicates; otherwise a vertical separation above 0.08 gives *above* or *below*, and failing that the larger horizontal axis gives *left_of*/*right_of* or *in_front_of*/*behind*. Pairs beyond a maximum range (0.75) emit no edge. The exported graph keeps both the predicate and the numeric displacement that produced it, so the same object identities, point sets, labels, and provenance can be reused by manual review or by a future learned relation head.

## 4 Experiments

We ask what helps when building open-vocabulary 3D scene graphs from a handful of RGB views, and how to measure it honestly. The second matters because an earlier version reported a sparse-view win for uncertainty-aware fusion that was a circular-evaluation artifact; we fix the protocol, then re-ask the first question on it. We describe the benchmark and protocol (Section 4.1), the independent reference that de-circularizes evaluation (Section 4.2), and the variants, metric, and implementation (Sections 4.3–4.5) before the results (Section 5).

### 4.1 Benchmark and Sparse-View Protocol

We evaluate on public TUM RGB-D indoor sequences [25], recognizable scenes with reproducible identifiers. The benchmark spans **28 object-rich sequences** across `freiburg1`–`freiburg3`: five paper-subset scenes

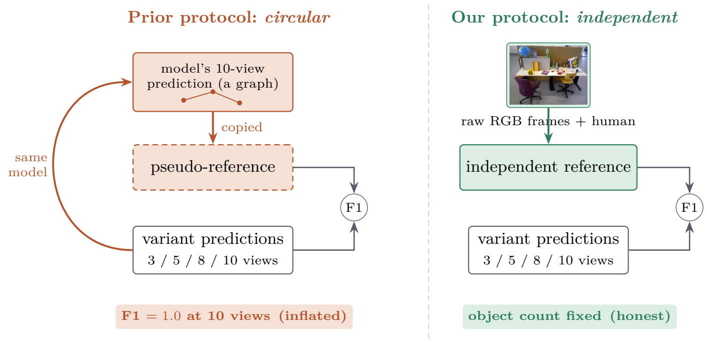


**Fig. 3.** De-circularizing the evaluation. **Left (prior):** the reference is *copied* from the model's own ten-view prediction, so the model is graded against itself: it scores 1.0 at ten views by construction, and a variant that splits more (the uncertainty gate) is rewarded for reproducing the reference's fragments. **Right (ours):** the reference is authored from the raw RGB frames, independent of any model output, so the object count is fixed and over-splitting is penalized. Both findings, the uncertainty gate reversing to a loss, duplicate-aware fusion winning, appear only under the independent reference.

(`freiburg1_room`, `freiburg1_desk`, `freiburg1_desk2`, `freiburg2_xyz`, `freiburg3_long_office_household`) carrying the human-verified gold reference, plus 23 further sequences released with the code. Each contributes ten RGB frames sampled at stride ten; the pipeline consumes only RGB. Per scene we run $N \in \{3, 5, 8, 10\}$ input views, a $28 \times 4$ grid of 112 graphs per variant, each produced end-to-end by the pipeline of Section 3. Reporting accuracy against $N$ reads sparse-view behavior directly rather than collapsing it to one dense-view number.

### 4.2 An Independently-Authored Reference

TUM RGB-D has no ground-truth open-vocabulary object inventory, so a reference must be constructed. The shortcut taken by the original version seeds the reference from the system's own dense-view prediction (its ten-view graph) and lightly checks it. This couples reference to predictor: a variant scores 1.0 at ten views *by construction*, and every sparser run is graded on how faithfully it reproduces that variant's own splitting pattern rather than on agreement with the scene (Figure 3, left), rewarding methods that leave proposals unmerged to reproduce the reference's fragments.

We instead author the reference *from the frames*. For each scene we enumerate an object multiset from the ten raw RGB frames alone, independent of the detection, descriptor, geometry, and fusion stack under test, then re-examine it in a second adversarial pass for missed or spurious objects. The five paper-subset scenes are additionally **human-verified** against the frames (the gold core); the other 23 use the two-pass draft without verification as scale corroboration, and both findings hold on the gold five alone. Since the inventory is fixed without reference to any model output, no variant can reach 1.0 at ten views and over-splitting is penalized. The reference is a scene-level multiset over all ten frames, so sparse-view predictions are recall-limited against it, depressing absolute F1 uniformly across variants and keeping the comparison fair.

### 4.3 Variants Compared

Holding the front-end (OWLv2 detections lifted by VGGT geometry) and the reference fixed, we compare seven variants that differ only in how per-view detections become 3D nodes and how those nodes are fused. **2d-only** pools per-view OWLv2 labels with no 3D lifting or fusion, an image-level upper reference on labels alone. **geometry-only** lifts detections to 3D and pools them with no semantic association. **semantic-lifting** associates lifted detections by appearance alone, without the geometric gate. **fixed-shrink** is an uncertainty-blind control that shrinks the fusion gate by a fixed, scene-agnostic factor, isolating whether the uncertainty *signal*, not merely a tighter gate, is what helps. **graph-fusion** is the cross-view fusion of Section 3.5 with no uncertainty modulation and no duplicate handling. **proposed** adds the rank-normalized uncertainty-aware gate (default $w$=0.3, swept over $\{0.1, 0.2, 0.3, 0.5, 0.8\}$) as an ablation. **graph-fusion-dedup (ours)** follows graph-fusion with the single duplicate-instance merge of Section 3.6: same-label nodes whose axis-aligned 3D boxes overlap (IoU $> \tau$, default $\tau$=0.1) are unioned and re-fused. Two pairings carry the argument: *fixed-shrink* versus *proposed* tests whether the uncertainty signal helps beyond an uninformed gate adjustment, and *graph-fusion* versus *graph-fusion-dedup* tests whether merging *more*, not less, is the productive direction.

The uncertainty-aware gate in PROPOSED assigns each lifted candidate an uncertainty

$$U_{v,k} = \lambda\,(1 - \bar{C}_{v,k}) + (1-\lambda)\,\mathrm{spread}(X_{v,k}), \qquad \lambda = 0.7, \tag{5}$$

combining the mean VGGT confidence $\bar{C}_{v,k}$ over the lifted points $X_{v,k}$ with a robust spread of those points (the 75th-percentile distance to their median, clipped to $[0, 1]$). These values are rank-normalized per scene; per candidate pair the joint uncertainty (the larger of the two) shrinks the centroid budget $\epsilon_g$ and raises the appearance bar $\epsilon_f$ in proportion to $w$, with a veto forbidding two highly uncertain nodes from seeding a merge. The gate can thus only *remove* merges, never add them, the wrong direction for an over-counting system; we keep it as an ablation, and the sweep over $w$ (Section 5) confirms the negative result.

### 4.4 Metric

Our primary metric is `object_label_f1`, the multiset precision, recall, and F1 of fused-node labels against the independent reference multiset, averaged over the 28 scenes (5 human-verified, 23 two-pass VLM-drafted). The multiset treatment makes the metric sensitive to over-counting: one physical monitor reported as four fused monitor nodes is penalized on precision, the failure mode the reframed contribution targets. We report per-scene wins, losses, and ties alongside the means, with significance from a two-sided sign test, and release the evaluation and aggregation code with the raw per-(scene, $N$, variant) scores.

### 4.5 Implementation Details

All heavy models are frozen and run off-the-shelf, with no per-scene optimization or fine-tuning. The detector is the OWLv2-base patch-16 ensemble checkpoint [17], successor to OWL-ViT [18], with score threshold 0.2 and per-class non-maximum suppression at box-IoU 0.5; detection boxes prompt SAM [11] for masks. Geometry is the feed-forward VGGT transformer [28] consuming RGB frames only, and cross-view association uses CLIP [22] and DINOv2 [19] descriptors. VGGT runs on GPU; detection, lifting, fusion, and evaluation run on CPU. All thresholds are fixed across scenes and view counts, and the duplicate-aware merge uses a single global IoU threshold $\tau$=0.1 shown robust across $\tau \in [0, 0.2]$ (Section 5).

### 4.6 Static and Dynamic Subsets

Six of the 28 sequences are *dynamic*, with a moving person (`desk_with_person`, `sitting_*`, `walking_*`), violating VGGT's static multi-view assumption; the other 22 are static. We report the subsets separately and analyze the duplicate-aware gain on each with the results (Section 5).

## 5 Results

All numbers are object-label F1 against the independent 28-scene reference of Section 4 (multiset precision, recall, and F1 of fused-node labels, mean over scenes at 3/5/8/10 input views). Section 5.5 contrasts this with the earlier prediction-seeded pseudo-reference.

### 5.1 Main Comparison

Table 1 compares the seven variants of Section 4.3. On real, detector-grounded labels GRAPH-FUSION is the strongest prior fusion variant, yet object-label F1 *falls* as views are added for every cross-view fusion variant: more views supply more detections of the same physical object, which fusion does not collapse, so precision degrades against a fixed reference. The non-fusion references 2D-ONLY and GEOMETRY-ONLY plateau. GRAPH-FUSION-DEDUP is the only variant whose accuracy *rises* toward the dense end, and is best at every view count (Figure 4).

**Table 1.** Object-label F1 on the independent reference (mean over 28 TUM RGB-D sequences, full multiset). FIXED-SHRINK is an uninformed control and PROPOSED the rank-normalized uncertainty-aware variant ($w$=0.3). Best per column in bold.

| Variant | v3 | v5 | v8 | v10 |
|---|---|---|---|---|
| 2D-ONLY | 0.5120 | 0.5003 | 0.4867 | 0.4753 |
| GEOMETRY-ONLY | 0.4617 | 0.4605 | 0.4594 | 0.4493 |
| SEMANTIC-LIFTING | 0.3752 | 0.2881 | 0.2123 | 0.1797 |
| FIXED-SHRINK (control) | 0.5048 | 0.4715 | 0.4448 | 0.4189 |
| GRAPH-FUSION (baseline) | 0.5129 | 0.4895 | 0.4664 | 0.4502 |
| PROPOSED (uncertainty, $w$=0.3) | 0.4801 | 0.4370 | 0.3841 | 0.3544 |
| **graph-fusion-dedup (ours)** | **0.5696** | **0.5723** | **0.5860** | **0.5725** |

### 5.2 Duplicate-Aware Fusion: the Positive Result

Duplicate-aware fusion adds a single post-fusion step: same-label fused nodes whose axis-aligned 3D boxes overlap (IoU $> \tau$, default $\tau$=0.1) are merged and re-fused, collapsing the duplicate nodes of one physical object into a single object (Figure 5). Dedup raises object-label F1 at every view count, by +0.057 to +0.122 over GRAPH-FUSION and by +0.065 to +0.154 over the uninformed FIXED-SHRINK control, and the gain *grows* with views, as expected for a step removing the duplicate detections that accumulate as more frames are seen. It wins on 22 to 27 of the 28 scenes at every view count (two-sided sign test $p < 0.001$; per-view deltas and per-scene W/L/T in Table S4, supplementary). The improvement holds on the 22 static scenes (+0.082 to +0.171), the 6 dynamic scenes with a moving person (+0.060 to +0.116), and the 5 human-verified gold scenes alone ($+0.068/+0.085/+0.142/+0.146$), so it is not an artifact of the VLM-drafted references. The gain is robust to the merge threshold, holding in the +0.075 to +0.118 range (average $\Delta$) across $\tau \in [0, 0.2]$ and fading only for strict $\tau$ (+0.030 at $\tau \geq 0.5$); a centroid-distance term does not help, so 3D-box IoU alone suffices (Fig. S2, supplementary).

### 5.3 Uncertainty-Aware Fusion: the Negative Result

We treat the uncertainty-aware fusion gate, the headline of an earlier framing of this work, as an ablation (full numbers in Table S3, supplementary). The recalibrated PROPOSED variant is *worse* at every view count (−0.033 to −0.096 vs. GRAPH-FUSION) and loses on essentially every scene (26 to 27 of 28 at v5–v10, sign test $p < 0.001$). A sweep over the uncertainty weight $w \in \{0.1, 0.2, 0.3, 0.5, 0.8\}$ does not rescue it: *no* weight in $[0.1, 0.8]$ beats GRAPH-FUSION at any view count, with larger weights worse at the dense end. The result survives reference filtering: dropping `unknown object` (which the open-vocabulary detector never predicts) or the "stuff" classes (wall/floor/ceiling) raises every variant's F1 but leaves the ranking unchanged.

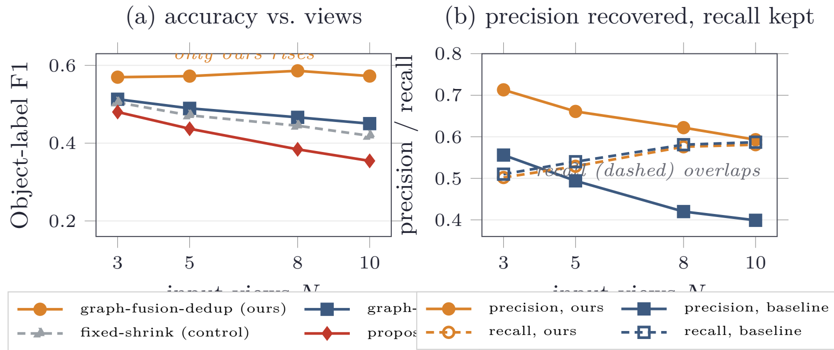


**Fig. 4.** Why duplicate-aware fusion wins, on the independent 28-scene reference. **(a)** Object-label F1 vs. number of input views: every prior cross-view fusion variant *degrades* as views are added (more frames yield more un-collapsed duplicates), while duplicate-aware fusion (amber) is the only variant whose accuracy *rises* and is best at every view count; the uncertainty gate (PROPOSED) sits below even the uninformed control. **(b)** The mechanism: dedup *recovers precision* (solid amber far above the baseline, $0.71 \to 0.59$ vs. $0.56 \to 0.40$) while leaving *recall essentially unchanged* (dashed lines overlap). Values match Table 1.

The negative result is informative, not a mere null: the gate does what it was designed to do, lifting recall (mean PROPOSED 0.516/0.556/0.596/0.602 vs. GRAPH-FUSION 0.510/0.540/0.581/0.587) but fragmenting correct objects and costing more precision than it buys (0.478/0.388/0.299/0.266 vs. 0.556/0.494/0.420/0.399), so net F1 is lower. This is the *opposite* axis to duplicate-aware fusion, which merges *more* where the system over-splits (Section 6).

### 5.4 Per-Class Characterization: the Over-Counting Signature

Per-class micro-averages (Tables S1 and S2, supplementary; dropping `unknown object`, which the detector never emits) localize the effect. For GRAPH-FUSION recall rises from 0.700 at three views to 0.802 at ten while precision falls from 0.519 to 0.331, so F1 peaks at the sparse end (0.596 at v3, 0.468 at v10): more frames find more objects but emit more duplicates, and the precision loss dominates. It concentrates in classes *found reliably but over-counted* (recall near 1.0, precision collapsing), such as `desk` (0.15), `floor` (0.27), `monitor` (0.43), and `chair` (0.45), whereas distinct, well-separated objects (`book`, `bottle`, `cup`) fuse cleanly.

Dedup at v8 confirms the mechanism, recovering precision on exactly the over-counted classes (`desk` $0.15 \to 0.45$, `floor` $0.27 \to 0.71$, `monitor` $0.43 \to 0.64$, `chair` $0.45 \to 0.71$) while leaving recall unchanged for almost every class; its only recall cost is $-0.04$ on closely-packed same-class instances (`desk`, `bottle`), which the precision gain dwarfs. The dominant error is over-counting of correctly-found objects, not mislabeling, precisely what duplicate-aware fusion targets;

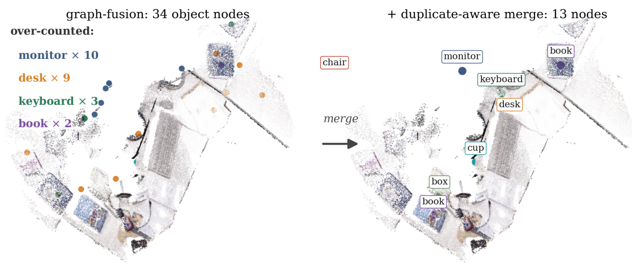


**Fig. 5.** Duplicate-aware fusion on the real `freiburg1_desk` cloud (ten views; desaturated so the object nodes stand out). **Left:** GRAPH-FUSION leaves the single monitor as ten nodes and the desk as nine, the over-counting that dominates the precision loss. **Right:** the merge collapses same-label overlapping nodes (34 → 13 nodes) into a clean labeled graph. (Shown by 3D proximity here; the metric uses 3D-box IoU, Section 4.3.)

separating closely-packed instances by appearance is left to future work (Section 6).

### 5.5 De-Circularization: the Earlier Pseudo-Reference Was Misleading

Both findings are visible only under an independent reference. Seeding the reference from the model's own ten-view GRAPH-FUSION output produces two artifacts: GRAPH-FUSION scores 1.0 at ten views *by construction* (the reference is its own prediction), and the uncertainty variant, which splits more, reproduces more of the reference's fragments and looks like a sparse-view *win* (+0.040/+0.021 at v3/v5). Fixing the object count independently reverses the uncertainty result and removes the dense-view 1.0 artifact (Tables S1 and S3, supplementary). A prediction-seeded reference is an easy, silent way to inflate sparse-view 3D scores, and it masked both findings.

## 6 Discussion

The two findings are one mechanism seen twice. The lifted system's dominant error is *over-counting*: one physical object survives cross-view fusion as several same-label nodes (Table S1, supplementary). As views are added, recall rises but precision falls faster, so accuracy peaks at the sparse end. The fix for this over-*splitting* is to *merge*: duplicate-aware fusion merges same-label nodes whose 3D boxes overlap, recovering precision while keeping recall (Table S2, supplementary), and it alone gains accuracy with more views, as added views then supply redundancy not fragmentation. The uncertainty-aware gate instead *tightens* merging where the system already over-splits: no weight in the sweep $w \in [0.1, 0.8]$ beats the non-gated baseline, and the default regresses by $-0.033$ to $-0.096$ across the four view counts (Table S3, supplementary). This is not a tuning failure but the predictable cost of gating merges where the error is too few merges, the clearest evidence for the over-counting diagnosis.

### 6.1 The Circular-Evaluation Lesson

This mechanism was invisible under a prediction-seeded reference. Seeded from the model's own dense-view output, it scores a method that reproduces its ten-view graph 1.0 *by construction* and rewards recall regardless of real objects, so the gate looked like a sparse-view win; it measures agreement with that model rather than recognition. Re-grading the identical outputs against an independently authored reference reverses that result and exposes the precision collapse. An easy trap in sparse-view 3D where dense ground truth is costly [6,8], this leads us to treat the independent reference and the documented pitfall as a first-class contribution and to recommend reporting any prediction-seeded reference alongside an independent one.

### 6.2 Limitations

Three limitations bound these results. First, of the 28 scenes only 5 are human-verified and 23 carry two-pass VLM-drafted references; both findings reach $p < 0.001$ over the 28 scenes and hold on the gold five alone (Section 5.2), so they do not rest on the drafted labels, though verifying the 23 remains future work. Second, box-IoU merging cannot separate distinct same-class objects whose axis-aligned 3D boxes overlap, such as two monitors on one desk, the merge's only recall cost ($-0.04$ on `desk` and `bottle` at eight views, Table S2, supplementary), which the precision gain dwarfs. Third, the pipeline assumes a static, "thing"-centric scene: VGGT's geometry degrades on dynamic sequences, "stuff" classes (`wall`, `floor`, `ceiling`) are treated as countable objects, and centroid-and-distance spatial rules cannot express the support, containment, or functional relations of richer 3D scene graphs [1,27,5].

## 7 Conclusion

We presented a detector-grounded open-vocabulary 3D scene-graph system, every heavy model frozen and no per-scene optimization, in which OWLv2 proposes labeled boxes, feed-forward VGGT geometry lifts them to 3D, and cross-view fusion assembles an inspectable object–relation graph from a handful of casual RGB views. On an independent TUM RGB-D benchmark its dominant error is over-counting, which a one-line duplicate-aware merge corrects, improving object-label F1 by 0.06 to 0.12 at every view count and winning on 22 to 27 of the 28 scenes ($p < 0.001$), while a recalibrated, fully swept uncertainty-aware gate gives no benefit, a negative result a prediction-seeded reference had disguised as a win. The de-circularized reference and the documented pitfall are the methodological core, and the dataset-agnostic benchmark and protocol are released. The next step follows from the merge's one failure mode: an appearance-based split separating closely-packed same-class instances within a merged group rather than a gate that blocks fusion.

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